02_Household_Forecast

```r
knitr::opts_chunk$set(echo = TRUE, warning = FALSE, message = FALSE)

library(knitr)
```

```
## Warning: package 'knitr' was built under R version 4.5.2
```

```r
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.5.2

## Warning: package 'tibble' was built under R version 4.5.2

## Warning: package 'tidyr' was built under R version 4.5.2

## Warning: package 'purrr' was built under R version 4.5.2

## Warning: package 'stringr' was built under R version 4.5.2

## Warning: package 'forcats' was built under R version 4.5.2

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.2     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(forecast)
library(lubridate)
library(ggplot2)
library(tseries)
```

```
## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```r
source("../utils/one_step_model.R")
```

Load Data

We use the train/test splits and differenced training data saved from the EDA notebook directly.

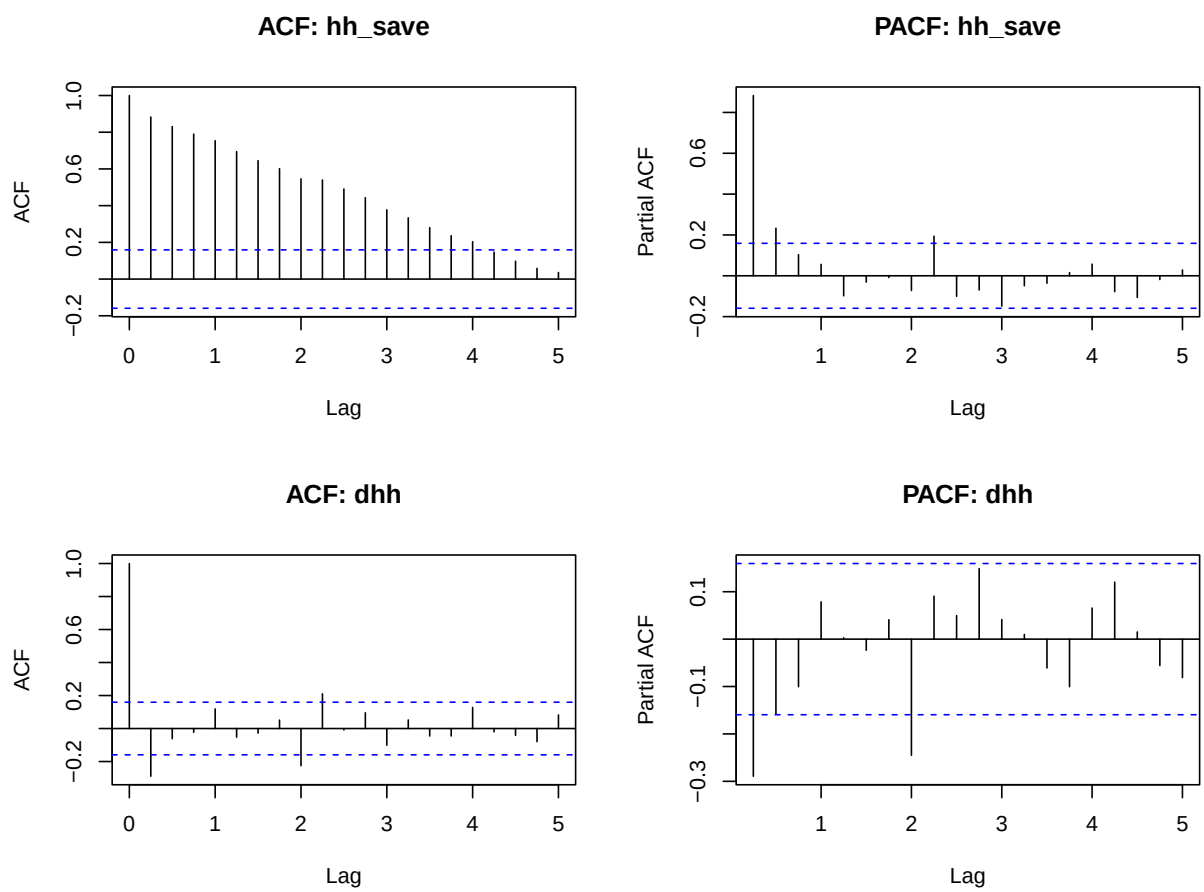
```r
train_df   <- readRDS("../output/train_df.rds")
test_df    <- readRDS("../output/test_df.rds")
macro_diff <- readRDS("../output/macro_df_diff.rds")

hh_train <- ts(train_df$hh_save, start = c(1977, 1), frequency = 4)
hh_test  <- ts(test_df$hh_save,  start = c(2015, 1), frequency = 4)
```

ARIMA Order Identification

```
dhh_train <- diff(hh_train)

par(mfrow = c(2, 2))
acf(hh_train, lag.max = 20, main = "ACF: hh_save")
pacf(hh_train, lag.max = 20, main = "PACF: hh_save")
acf(dhh_train, lag.max = 20, main = "ACF: dhh")
pacf(dhh_train, lag.max = 20, main = "PACF: dhh")
```




```
par(mfrow = c(1, 1))
```

- The ACF of `hh_save` decays slowly, confirming non-stationarity and the need for differencing.
- After differencing, the ACF/PACF of `dhh` guides the choice of p and q for ARIMA.
- The candidate models we are considering are
  - ARIMA(1, 1, 1), ARIMA(0, 1, 1), ARIMA(2,1,0)
  - SARIMA(1, 1, 1)(1, 0, 0)[4], SARIMA(1, 1, 1)(0, 0, 1)[4], etc.

One-Step Benchmark Models

All benchmarks forecast `hh_save` at level directly.

```r
rmse_tbl <- tibble(Model = character(), RMSE = double())
fc_store <- list()

# Average
res <- iid_fc(hh_train, hh_test)
rmse_tbl <- add_row(rmse_tbl, Model = "Average", RMSE = res$rmse)
fc_store$Average <- res$fc

print(res$rmse)
```

IID (Average Forecasts)

```
## [1] 18434.48
```

```r
# Average by Year
res <- iidbyyear_fc(hh_train, hh_test, periodicity = 4)
rmse_tbl <- add_row(rmse_tbl, Model = "Average by Year", RMSE = res$rmse)
fc_store$`Average by Year` <- res$fc
print(res$rmse)
```

```
## [1] 20531.04
```

```r
# Average by Period
res <- iidbyperiod_fc(hh_train, hh_test, periodicity = 4)
rmse_tbl <- add_row(rmse_tbl, Model = "Average by Period", RMSE = res$rmse)
fc_store$`Average by Period` <- res$fc
print(res$rmse)
```

```
## [1] 18474.54
```

Persistence

```r
# Persistence forecast
res <- persist_fc(hh_train, hh_test)
rmse_tbl <- add_row(rmse_tbl, Model = "Persistence", RMSE = res$rmse)
fc_store$Persistence <- res$fc

print(res$rmse)
```

```
## [1] 13036.82
```

Holt-Winters Variants

SES

```r
# Simple Exponential Smoothing (SES)
hw_ses <- HoltWinters(hh_train, beta = FALSE, gamma = FALSE)
res <- esm_fc(hh_train, hh_test,
              alpha = hw_ses$alpha,
              level = as.numeric(hw_ses$coefficients[1]))
rmse_tbl <- add_row(rmse_tbl, Model = "SES", RMSE = res$rmse)
fc_store$SES <- res$fc

print(res$rmse)
```

```
## [1] 13503.72
```

Holt Linear

```
hw_holt <- HoltWinters(hh_train, gamma = FALSE)
res <- lholt_fc(hh_train, hh_test,
                alpha = hw_holt$alpha,
                beta  = hw_holt$beta,
                level = as.numeric(hw_holt$coefficients[1]),
                slope = as.numeric(hw_holt$coefficients[2]))
rmse_tbl <- add_row(rmse_tbl, Model = "Holt Linear", RMSE = res$rmse)
fc_store$`Holt Linear` <- res$fc

print(res$rmse)
```

```
##    alpha
## 13881.24
```

Holt-Winters Seasonal Additive

```
hw_add <- HoltWinters(hh_train, seasonal = "additive")
res <- aseason_fc(hh_train, hh_test,
                  alpha  = hw_add$alpha,
                  beta   = hw_add$beta,
                  gamma  = hw_add$gamma,
                  level  = as.numeric(hw_add$coefficients[1]),
                  slope  = as.numeric(hw_add$coefficients[2]),
                  season = as.numeric(hw_add$coefficients[3:6]))

rmse_tbl <- add_row(rmse_tbl, Model = "HW Additive", RMSE = res$rmse)
fc_store$`HW Additive` <- res$fc

print(res$rmse)
```

```
## alpha
## 14080
```

ARIMA One-Step Forecast

```
# Safely extract ARIMA order
extract_order <- function(ord, keys, default = 0L) {
  vals <- ord[keys]
  vals[is.na(vals)] <- default
  as.integer(vals)
}

fit_arima <- auto.arima(hh_train,
                        seasonal = TRUE,
                        stepwise = FALSE,
                        approximation = FALSE)
# Print Auto ARIMA summary
summary(fit_arima)
```

```
## Series: hh_train
## ARIMA(0,1,1)(2,0,0)[4]
##
```

```
## Coefficients:
##           ma1    sar1     sar2
##       -0.3196  0.1262  -0.1870
## s.e.   0.0838  0.0803   0.0807
##
## sigma^2 = 37418227:  log likelihood = -1529.51
## AIC=3067.01   AICc=3067.29   BIC=3079.08
##
## Training set error measures:
##                    ME    RMSE      MAE       MPE     MAPE      MASE       ACF1
## Training set 274.6611 6036.02 4569.573 -6.008478 20.27408 0.6084367 0.01396942
```

```r
# Try alternative ARIMA specifications to check robustness of order selection
fit_arima_111 <- arima_one_step_fc(
  train = hh_train,
  holdout = hh_test,
  order = c(1, 1, 1),
  include.mean = TRUE
)

fit_arima_011 <- arima_one_step_fc(
  train = hh_train,
  holdout = hh_test,
  order = c(0, 1, 1),
  include.mean = TRUE
)

fit_arima_210 <- arima_one_step_fc(
  train = hh_train,
  holdout = hh_test,
  order = c(2, 1, 0),
  include.mean = TRUE
)

fit_arima_110 <- arima_one_step_fc(
  train = hh_train,
  holdout = hh_test,
  order = c(1, 1, 0),
  include.mean = TRUE
)

arima_tbl <- tibble(
  Model = c("ARIMA(1,1,1)", "ARIMA(0,1,1)", "ARIMA(2,1,0)", "ARIMA(1,1,0)"),
  RMSE = c(fit_arima_111$rmse, fit_arima_011$rmse,
           fit_arima_210$rmse, fit_arima_110$rmse)
)

arima_tbl
```

```
## # A tibble: 4 x 2
##   Model          RMSE
##   <chr>         <dbl>
## 1 ARIMA(1,1,1) 13310.
## 2 ARIMA(0,1,1) 13499.
## 3 ARIMA(2,1,0) 13621.
```

```
## 4 ARIMA(1,1,0) 13567.
```

```r
arima_fcs <- list(
  "ARIMA(1,1,1)" = fit_arima_111$fc,
  "ARIMA(0,1,1)" = fit_arima_011$fc,
  "ARIMA(2,1,0)" = fit_arima_210$fc,
  "ARIMA(1,1,0)" = fit_arima_110$fc
)
best_arima_idx   <- which.min(arima_tbl$RMSE)
best_arima_label <- arima_tbl$Model[best_arima_idx]
rmse_tbl <- add_row(rmse_tbl, Model = best_arima_label, RMSE = arima_tbl$RMSE[best_arima_idx])
fc_store[[best_arima_label]] <- arima_fcs[[best_arima_label]]
```

```r
ord <- forecast::arimaorder(fit_arima)
res <- sarima_one_step_fc(
  train    = hh_train,
  holdout  = hh_test,
  order    = extract_order(ord, c("p", "d", "q")),
  seasonal = extract_order(ord, c("P", "D", "Q")),
  include.mean = TRUE
)
rmse_tbl <- add_row(rmse_tbl, Model = "Auto ARIMA", RMSE = res$rmse)
fc_store$`Auto ARIMA` <- res$fc
print(res$rmse)
```

```
## [1] 14092.74
```

```r
fit_sarima_111_100 <- sarima_one_step_fc(
  train = hh_train,
  holdout = hh_test,
  order = c(1, 1, 1),
  seasonal = c(1, 0, 0),
  include.mean = TRUE
)

fit_sarima_011_100 <- sarima_one_step_fc(
  train = hh_train,
  holdout = hh_test,
  order = c(0, 1, 1),
  seasonal = c(1, 0, 0),
  include.mean = TRUE
)

fit_sarima_111_101 <- sarima_one_step_fc(
  train = hh_train,
  holdout = hh_test,
  order = c(1, 1, 1),
  seasonal = c(1, 0, 1),
  include.mean = TRUE
)

fit_sarima_011_101 <- sarima_one_step_fc(
  train = hh_train,
  holdout = hh_test,
  order = c(0, 1, 1),
```

```r
  seasonal = c(1, 0, 1),
  include.mean = TRUE
)

fit_sarima_111_001 <- sarima_one_step_fc(
  train = hh_train,
  holdout = hh_test,
  order = c(1, 1, 1),
  seasonal = c(0, 0, 1),
  include.mean = TRUE
)

fit_sarima_011_001 <- sarima_one_step_fc(
  train = hh_train,
  holdout = hh_test,
  order = c(0, 1, 1),
  seasonal = c(0, 0, 1),
  include.mean = TRUE
)

fit_sarima_111_002 <- sarima_one_step_fc(
  train = hh_train,
  holdout = hh_test,
  order = c(1, 1, 1),
  seasonal = c(0, 0, 2),
  include.mean = TRUE
)

fit_sarima_011_002 <- sarima_one_step_fc(
  train = hh_train,
  holdout = hh_test,
  order = c(0, 1, 1),
  seasonal = c(0, 0, 2),
  include.mean = TRUE
)

sarima_tbl <- tibble(
  Model = c(
    "SARIMA(1,1,1)(1,0,0)[4]",
    "SARIMA(0,1,1)(1,0,0)[4]",
    "SARIMA(1,1,1)(1,0,1)[4]",
    "SARIMA(0,1,1)(1,0,1)[4]",
    "SARIMA(1,1,1)(0,0,1)[4]",
    "SARIMA(0,1,1)(0,0,1)[4]",
    "SARIMA(1,1,1)(0,0,2)[4]",
    "SARIMA(0,1,1)(0,0,2)[4]"
  ),
  RMSE = c(
    fit_sarima_111_100$rmse,
    fit_sarima_011_100$rmse,
    fit_sarima_111_101$rmse,
    fit_sarima_011_101$rmse,
    fit_sarima_111_001$rmse,
```

```r
    fit_sarima_011_001$rmse,
    fit_sarima_111_002$rmse,
    fit_sarima_011_002$rmse
  )
)

sarima_tbl
```

```
## # A tibble: 8 x 2
##   Model                      RMSE
##   <chr>                     <dbl>
## 1 SARIMA(1,1,1)(1,0,0)[4] 13434.
## 2 SARIMA(0,1,1)(1,0,0)[4] 13699.
## 3 SARIMA(1,1,1)(1,0,1)[4] 13677.
## 4 SARIMA(0,1,1)(1,0,1)[4] 13995.
## 5 SARIMA(1,1,1)(0,0,1)[4] 13547.
## 6 SARIMA(0,1,1)(0,0,1)[4] 13870.
## 7 SARIMA(1,1,1)(0,0,2)[4] 13774.
## 8 SARIMA(0,1,1)(0,0,2)[4] 14023.
```

```r
sarima_fcs <- list(
  "SARIMA(1,1,1)(1,0,0)[4]" = fit_sarima_111_100$fc,
  "SARIMA(0,1,1)(1,0,0)[4]" = fit_sarima_011_100$fc,
  "SARIMA(1,1,1)(1,0,1)[4]" = fit_sarima_111_101$fc,
  "SARIMA(0,1,1)(1,0,1)[4]" = fit_sarima_011_101$fc,
  "SARIMA(1,1,1)(0,0,1)[4]" = fit_sarima_111_001$fc,
  "SARIMA(0,1,1)(0,0,1)[4]" = fit_sarima_011_001$fc,
  "SARIMA(1,1,1)(0,0,2)[4]" = fit_sarima_111_002$fc,
  "SARIMA(0,1,1)(0,0,2)[4]" = fit_sarima_011_002$fc
)
best_sarima_idx   <- which.min(sarima_tbl$RMSE)
best_sarima_label <- sarima_tbl$Model[best_sarima_idx]
rmse_tbl <- add_row(rmse_tbl, Model = best_sarima_label, RMSE = sarima_tbl$RMSE[best_sarima_idx])
fc_store[[best_sarima_label]] <- sarima_fcs[[best_sarima_label]]
```

ARIMAX One-Step Forecast

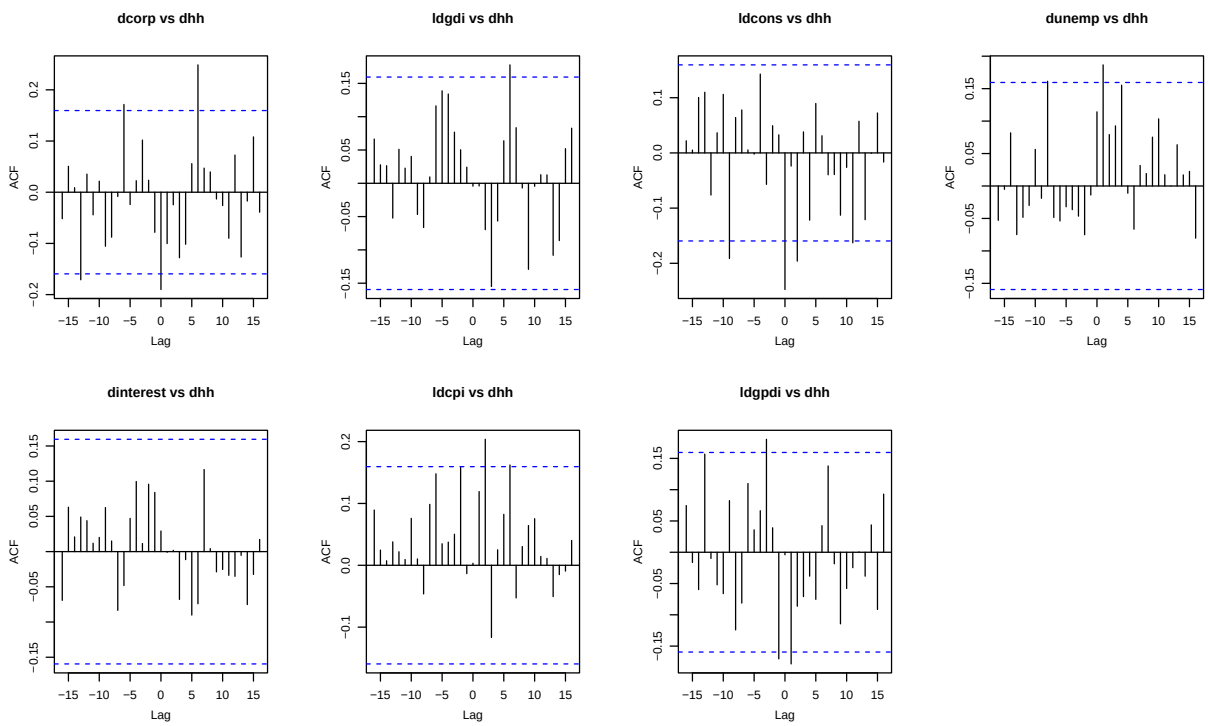
Exogenous Variable Selection via CCF

We use the differenced training data from the EDA to inspect cross-correlations with `dhh`. A variable is selected for ARIMAX only if it shows significant correlation at **negative lags** (the regressor leads `dhh`), to avoid overfitting with only 20 holdout observations.

```r
candidates <- c("dcorp", "ldgdi", "ldcons", "dunemp", "dinterest", "ldcpi", "ldgpdi")

par(
  mfrow = c(2, 4),
  mar = c(5, 5, 3, 1),
  mgp = c(2, 0.8, 0),
  cex.main = 0.95,
  cex.lab = 0.9,
  cex.axis = 0.85
)
for (v in candidates) {
  ccf(na.omit(macro_diff[[v]]), na.omit(macro_diff$dhh),
```

```r
      lag.max = 16,
      main = paste(v, "vs dhh"))
}
par(mfrow = c(1, 1))
```




```r
ccf_values <- sapply(candidates, function(v) {
  ccf_res <- ccf(na.omit(macro_diff[[v]]), na.omit(macro_diff$dhh), lag.max = 16, plot = FALSE)
  max_ccf <- max(abs(ccf_res$acf[ccf_res$lag < 0]))
  return(max_ccf)
})

# Identify the lag with the maximum absolute CCF for each candidate
ccf_lags <- sapply(candidates, function(v) {
  ccf_res <- ccf(na.omit(macro_diff[[v]]), na.omit(macro_diff$dhh), lag.max = 16, plot = FALSE)
  max_idx <- which.max(abs(ccf_res$acf[ccf_res$lag < 0]))
  return(ccf_res$lag[max_idx])
})

ccf_tbl <- tibble(
  Variable = candidates,
  Max_CCF = ccf_values,
  Lag_at_Max_CCF = ccf_lags
)

ccf_tbl
```

```
## # A tibble: 7 x 3
##   Variable  Max_CCF Lag_at_Max_CCF
##   <chr>       <dbl>          <dbl>
## 1 dcorp      0.171              -6
```

```
## 2 ldgdi      0.139              -5
## 3 ldcons     0.191              -9
## 4 dunemp     0.161              -8
## 5 dinterest  0.0996             -4
## 6 ldcpi      0.158              -2
## 7 ldgpdi     0.181              -3
```

Based on the CCF evidence and the EDA findings, we select regressors with clear leading relationship:

- ldgpdi
- ldcpi

```r
# Compute stationary exogenous variables across the full sample
arimax_df <- bind_rows(train_df, test_df) |>
  arrange(date) |>
  mutate(
    ldgpdi  = log(gpdi) - log(lag(gpdi)),
    ldcpi = log1p(cpi) - log1p(lag(cpi)),
    ldcpi_lag2 = lag(ldcpi, 2),
    ldgpdi_lag1 = lag(ldgpdi, 1),
    ldgpdi_lag3 = lag(ldgpdi, 3)
  ) |>
  drop_na(ldgpdi_lag1, ldcpi_lag2, ldgpdi_lag3) # Drop initial rows with NA due to differencing and lag

arimax_vars <- c("ldgpdi_lag1", "ldcpi_lag2", "ldgpdi_lag3")

x_train <- arimax_df |> filter(date <= as.Date("2014-12-31")) |>
  select(all_of(arimax_vars)) |>
  as.matrix()
x_test <- arimax_df |> filter(date > as.Date("2014-12-31")) |>
  select(all_of(arimax_vars)) |>
  as.matrix()
```

Fitting ARIMAX models

```r
# Align hh_save accordingly
hh_train_x <- ts(
  arimax_df |> filter(date <= as.Date("2014-12-31")) |> pull(hh_save),
  start = c(
    year(min(arimax_df$date)),
    quarter(min(arimax_df$date))
  ), frequency = 4
)

# Fit ARIMAX with selected regressors
fit_arimax <- auto.arima(hh_train_x,
                         xreg = x_train,
                         seasonal = FALSE,
                         stepwise = FALSE,
                         approximation = FALSE)
summary(fit_arimax)
```

```
## Series: hh_train_x
## Regression with ARIMA(2,1,3) errors
##
```

```
## Coefficients:
##          ar1      ar2      ma1     ma2      ma3  ldgpdi_lag1  ldcpi_lag2
##       0.8564  -0.8167  -1.2608  1.1515  -0.2074    -28461.19    61828.28
## s.e.  0.1240   0.1142   0.1613  0.2080   0.1337      9475.43    65105.12
##       ldgpdi_lag3
##          4279.895
## s.e.     9725.984
##
## sigma^2 = 36384042:  log likelihood = -1484.58
## AIC=2987.15   AICc=2988.47   BIC=3014.07
##
## Training set error measures:
##                     ME     RMSE      MAE       MPE    MAPE      MASE        ACF1
## Training set 209.2318 5845.639 4430.037 -6.176887 19.5516 0.5810266 0.002096567
```

```r
arimax_213 <- arimax_one_step_fc(
  train = hh_train_x,
  holdout = hh_test,
  order = c(2, 1, 3),
  xreg_train = x_train,
  xreg_holdout = x_test,
  include.mean = TRUE
)

arimax_212 <- arimax_one_step_fc(
  train = hh_train_x,
  holdout = hh_test,
  order = c(2, 1, 2),
  xreg_train = x_train,
  xreg_holdout = x_test,
  include.mean = TRUE
)

arimax_112 <- arimax_one_step_fc(
  train = hh_train_x,
  holdout = hh_test,
  order = c(1, 1, 2),
  xreg_train = x_train,
  xreg_holdout = x_test,
  include.mean = TRUE
)

arimax_111 <- arimax_one_step_fc(
  train = hh_train_x,
  holdout = hh_test,
  order = c(1, 1, 1),
  xreg_train = x_train,
  xreg_holdout = x_test,
  include.mean = TRUE
)

arimax_011 <- arimax_one_step_fc(
  train = hh_train_x,
  holdout = hh_test,
```

```r
  order = c(0, 1, 1),
  xreg_train = x_train,
  xreg_holdout = x_test,
  include.mean = TRUE
)

arimax_tbl <- tibble(
  Model = c(
    "ARIMAX(2,1,3)",
    "ARIMAX(2,1,2)",
    "ARIMAX(1,1,2)",
    "ARIMAX(1,1,1)",
    "ARIMAX(0,1,1)"
  ),
  RMSE = c(
    arimax_213$rmse,
    arimax_212$rmse,
    arimax_112$rmse,
    arimax_111$rmse,
    arimax_011$rmse
  )
)

arimax_tbl
```

```
## # A tibble: 5 x 2
##   Model           RMSE
##   <chr>          <dbl>
## 1 ARIMAX(2,1,3) 13484.
## 2 ARIMAX(2,1,2) 13595.
## 3 ARIMAX(1,1,2) 13385.
## 4 ARIMAX(1,1,1) 13384.
## 5 ARIMAX(0,1,1) 13578.
```

Theoretical Model (Precautionary Saving)

In Macroeconomics, the precautionary saving motive is modeled by the permanent income model. Where, assuming a representative household is constant relative risk averse, we have:

$$C_t^{-\gamma} = \beta \mathbb{E}_t[(1 + r_{t+1})C_{t+1}]$$

where C_t is consumption, β is the discount factor, γ is the coefficient of relative risk aversion, and r_{t+1} is the real interest rate. This implies that consumption growth depends on expected returns and the intertemporal elasticity of substitution.

Taking the log-transformation and Taylor expansion will eventually give us

$$c_t = c_{t+1} - \frac{1}{\gamma} \ln \beta - \frac{1}{\gamma} r_{t+1} - \frac{\gamma}{2} \mathrm{Var}_t(\Delta c_{t+1}) - \varepsilon_{t+1}$$

Where with the identity $s_t = y_t - c_t$, the first difference of saving is captured by

$$\Delta s_t = \Delta y_t - \Delta \mathbb{E}_t[c_{t+1}] + \frac{1}{\gamma} \Delta r_{t+1} + \frac{\gamma}{2} \Delta \mathrm{Var}_t(\Delta c_{t+1}) + \varepsilon_{t+1}$$

This suggests we can fit the following empirical model

$$dhh_t = \alpha + \sum_{i=0}^{p} \phi_i dhh_{t-i} + \beta_1 dinterest_{t-1} + \beta_2 dunemp_{t-1} + \beta_3 ldgdi_{t-1} + \beta_4 ldgpdi_{t-1} + \varepsilon_t$$

Where we use the unemployment and Gross Private Domestic Investment (GPDI) as proxies for uncertainty, and the real interest rate as the return variable. We lag all regressors by 1 quarter to avoid look-ahead bias in forecasting.

```
pim_df <- bind_rows(train_df, test_df) |>
  arrange(date) |>
  mutate(
    dinterest = interest - lag(interest),
    dunemp = unemp - lag(unemp),
    ldgdi = log(gdi) - log(lag(gdi)),
    ldgpdi = log(gpdi) - log(lag(gpdi)),
    dinterest_lag1 = lag(dinterest, 1),
    dunemp_lag1 = lag(dunemp, 1),
    ldgdi_lag1 = lag(ldgdi, 1),
    ldgpdi_lag1 = lag(ldgpdi, 1)
  ) |>
  drop_na(ldgdi_lag1)

pim_train <- pim_df |> filter(date <= as.Date("2014-12-31"))|>
  select(dinterest_lag1, dunemp_lag1, ldgdi_lag1, ldgpdi_lag1)|>
  as.matrix()
pim_test <- pim_df |> filter(date > as.Date("2014-12-31"))|>
  select(dinterest_lag1, dunemp_lag1, ldgdi_lag1, ldgpdi_lag1)|>
  as.matrix()

hh_train_x <- ts(
  pim_df |> filter(date <= as.Date("2014-12-31"))|>pull(hh_save),
  start = c(
    year(min(pim_df$date)),
    quarter(min(pim_df$date))
  ),
  frequency = 4
)
```

```
pim_fit <- auto.arima(hh_train_x,
                      xreg = pim_train,
                      seasonal = FALSE,
                      stepwise = FALSE,
                      approximation = FALSE)
pim_fit
```

```
## Series: hh_train_x
## Regression with ARIMA(0,1,1) errors
##
## Coefficients:
##           ma1  dinterest_lag1  dunemp_lag1  ldgdi_lag1  ldgpdi_lag1
##       -0.4042         29.0394     779.5446    8971.981    -28498.08
## s.e.   0.0768        483.7370    1438.3098   54724.346     11894.54
##
## sigma^2 = 38033512:  log likelihood = -1509.29
## AIC=3030.58   AICc=3031.17   BIC=3048.6
```

```
pim_111 <- arimax_one_step_fc(
  train = hh_train_x,
  holdout = hh_test,
```

```r
  order = c(1, 1, 1),
  xreg_train = pim_train,
  xreg_holdout = pim_test,
  include.mean = TRUE
)

pim_011 <- arimax_one_step_fc(
  train = hh_train_x,
  holdout = hh_test,
  order = c(0, 1, 1),
  xreg_train = pim_train,
  xreg_holdout = pim_test,
  include.mean = TRUE
)


pim_110 <- arimax_one_step_fc(
  train = hh_train_x,
  holdout = hh_test,
  order = c(1, 1, 0),
  xreg_train = pim_train,
  xreg_holdout = pim_test,
  include.mean = TRUE
)

pim_012 <- arimax_one_step_fc(
  train = hh_train_x,
  holdout = hh_test,
  order = c(0, 1, 2),
  xreg_train = pim_train,
  xreg_holdout = pim_test,
  include.mean = TRUE
)

pim_tbl <- tibble(
  Model = c(
    "PIM ARIMAX(1,1,1)",
    "PIM ARIMAX(0,1,1)",
    "PIM ARIMAX(1,1,0)",
    "PIM ARIMAX(0,1,2)"
  ),
  RMSE = c(
    pim_111$rmse,
    pim_011$rmse,
    pim_110$rmse,
    pim_012$rmse
  )
)

pim_tbl
```

```
## # A tibble: 4 x 2
##   Model                RMSE
##   <chr>               <dbl>
```

```
## 1 PIM ARIMAX(1,1,1) 13526.
## 2 PIM ARIMAX(0,1,1) 13681.
## 3 PIM ARIMAX(1,1,0) 13749.
## 4 PIM ARIMAX(0,1,2) 13473.
```

```
pim_fcs <- list(
  "PIM ARIMAX(1,1,1)" = pim_111$fc,
  "PIM ARIMAX(0,1,1)" = pim_011$fc,
  "PIM ARIMAX(1,1,0)" = pim_110$fc,
  "PIM ARIMAX(0,1,2)" = pim_012$fc
)
best_pim_idx   <- which.min(pim_tbl$RMSE)
best_pim_label <- pim_tbl$Model[best_pim_idx]
rmse_tbl <- add_row(rmse_tbl, Model = best_pim_label, RMSE = pim_tbl$RMSE[best_pim_idx])
fc_store[[best_pim_label]] <- pim_fcs[[best_pim_label]]
```

RMSE Ranking

```
rmse_tbl <- rmse_tbl |>
  mutate(RMSE = round(RMSE, 4)) |>
  arrange(RMSE)

rmse_tbl
```

```
## # A tibble: 11 x 2
##    Model                      RMSE
##    <chr>                     <dbl>
##  1 Persistence             13037.
##  2 ARIMA(1,1,1)            13310.
##  3 SARIMA(1,1,1)(1,0,0)[4] 13434.
##  4 PIM ARIMAX(0,1,2)       13473.
##  5 SES                     13504.
##  6 Holt Linear             13881.
##  7 HW Additive             14080.
##  8 Auto ARIMA              14093.
##  9 Average                 18434.
## 10 Average by Period       18475.
## 11 Average by Year         20531.
```

```
best_model <- rmse_tbl$Model[1]
cat("Best model by one-step holdout RMSE:", best_model, "\n")
```

```
## Best model by one-step holdout RMSE: Persistence
```

Holdout Visualization

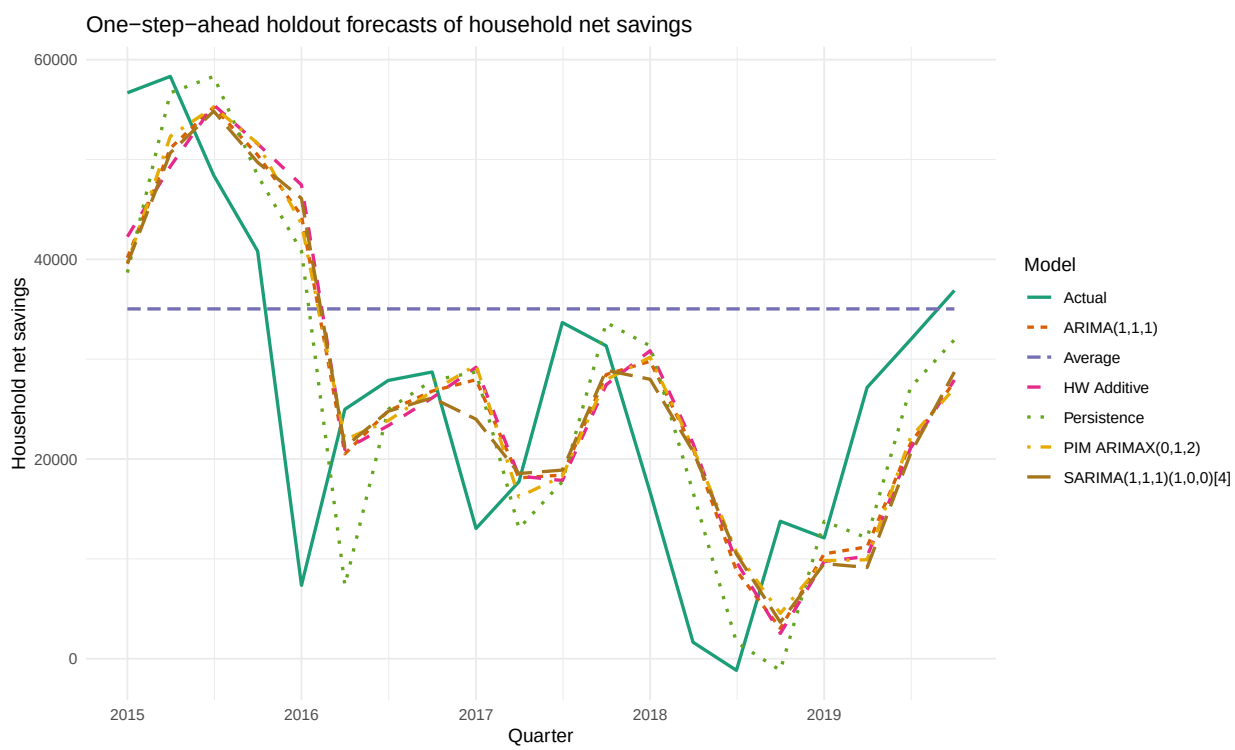
```
fc_df <- tibble(date = test_df$date, Actual = as.numeric(hh_test))


for (m in names(fc_store)) {
  fc_df[[m]] <- as.numeric(fc_store[[m]])
}

fc_long <- fc_df |>
```

```r
  pivot_longer(cols = -date, names_to = "Model", values_to = "value")|>
  filter(
    Model %in% c("Actual", "Average", "Persistence",
                 "HW Additive",
                 best_arima_label,
                 best_sarima_label,
                 best_pim_label)
  ) # Keep only benchmarks and the best HW and ARIMA/SARIMA models for clarity

ggplot(fc_long, aes(x = date, y = value, color = Model, linetype = Model)) +
  geom_line(linewidth = 0.9) +
  theme_minimal() +
  scale_color_brewer(palette = "Dark2") +
  labs(
    title = "One-step-ahead holdout forecasts of household net savings",
    x = "Quarter",
    y = "Household net savings",
    color = "Model",
    linetype = "Model"
  )
```



Interpretation

- All benchmarks and ARIMA/ARIMAX models forecast `hh_save` directly; differencing is handled internally by `auto.arima` via the d parameter.
- The ARIMAX uses only CCF-justified regressors (ldgpdi_lag1, ldcpi_lag2, ldgpdi_lag3) to avoid overfitting the small holdout window.
- The RMSE ranking table determines the best model for household net savings forecasting.